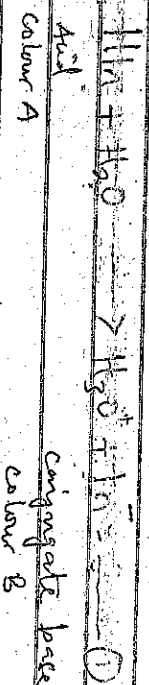


Acid-Base Titration

An acid-base titration is a laboratory procedure used in qualitative chemical analysis to determine the concentration of either a base or an acid. It is often carried out using a visual indicator to determine the end-point. An acid of known concentration is added from a burette to an alkaline solution (base) of unknown concentration in a conical flask. The end point occurs when stoichiometric amount of the acid has been added to the base, at this point complete neutralisation is achieved. The solution consists of salt and water only. From the values of the molarity of the acid, the volume of acid used and the volume of base used, the molarity of the base can be determined.

Indicators (Acid-Base)

An acid-base indicator is a laboratory indicator that changes colour when an acid or base is added. They are usually weak acids or bases. Acid-base indicators are also known as pH indicators. Consider an indicator which is a weak acid with general formula HIn . When they dissolved in water, an equilibrium is established between the weak acid & its conjugate base.



We can apply the equilibrium law to the above expression in general for a weak acid HIn , the dissociation constant is given by:

$$K_{In} = \frac{[H_3O^+][In^-]}{[HIn]} \quad (2)$$

where K_{In} is called the indicator dissociation constant. The colour of an indicator changes from A to B or vice versa at a point called the turning point, at this point the concentration of the acid HIn is equal to the concentration of the conjugate base In^- .

(10)

$$[H_3O^+] = [H^+] \quad (3)$$

Substituting the value of equation (3) into equation (2) we have

$$K_{1n} = \frac{[H_3O^+]}{[H^+]} \quad (4)$$

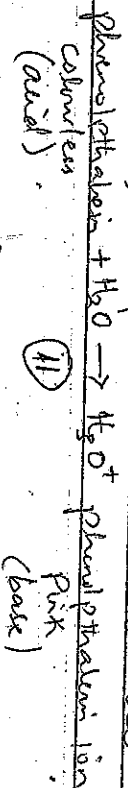
The pH at a turning point is called pK_{1n} . This is the point at which half the indicator is in acid form & the other half is in conjugate base.

Indicator Range

Acid-base indicator each have specific range within which the produce colour changes. At low pH, a weak acid is almost entirely in HIn (indicator) form, the colour of this form predominates & become observable. As pH increases, the intensity of colour "A" of the colour "A" diminishes. At equilibrium in equation (1) is pushed to the right. Consequently, intensity of colour "B" of the conjugate base increases. This observable colour changes from A to B. An indicator is almost effective if the distinct colour changes over a narrow range of pH value. For most indicators, a range is written (4 or 1) of the pK_{1n} value as shown in the table below.

Indicators	Acid colour	Base colour	pK_{1n}	pH Range
Thymol blue	Red	Yellow	1.5	1.2 - 2.8
Methyl orange	Red	Yellow	3.7	3.2 - 4.4
Bromocresol green	Yellow	Blue	4.7	3.8 - 5.4
Methyl red	Yellow	Red	5.1	4.8 - 6.0
Bromothymol blue	Yellow	Blue	7.0	6.0 - 7.6

Acid-base indicator has sharply different colours in its dissociation or undissociates state. Example, Phenolphthalein is a colourless weak acid which dissociates in water forming pink ion (anion). Under acidic condition, the equilibrium is pushed to the left. The concentration of the anion is too low for the pink colour to be observable. However, under alkaline condition, the equilibrium is pushed to the right & the concentration of the anion becomes sufficient for the pink colour to be observed.



Buffer Solution

A Buffer Solution is a solution that resist the change in pH when small amount of acid or base is added. It is a solution whose pH is not altered to any great extent by the addition of small quantities of either an acid or a base.

Characteristics of Buffer Solution

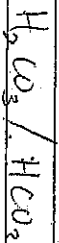
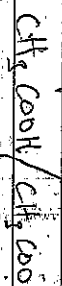
- i. It has a definite pH.
- ii. Its pH does not change on standing for long period of time.
- iii. Its pH does not change on dilution.
- iv. Its pH is slightly changed by the addition of small quantity of an acid or a base.

Applications of Buffer Solution

- i. Buffer are used in industrial processes such as manufacture of paper, dyes, inks, paints, drugs and so on.
- ii. Buffers are also employed in agriculture, drug products & preservation of various types of foods and fruits.
- iii. It is used to determine the pH with the help of indicators in buffers (bicarbonate buffer) is used to keep the pH of human body around a pH of 7.4.
- iv. For the removal of phosphate ion in the quantitative inorganic analysis.

* Buffers are solution of weak acid and the salt of its anions. This combination (i.e. weak acid & the salt) has the important property of resisting the pH (H^+ concentration) on any combination of weak acids & their salts are used as buffers to control pH in laboratory & in industries.

Examples

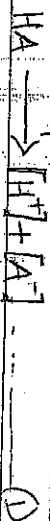


* Buffer Solutions are divided into four (4) categories

- i Strong acid buffers
- ii Strong base buffers
- iii Weak acid buffers
- iv Weak base buffers

(i) Weak acid Buffers

For a weak acid HA, the equilibrium between the acid & salt can be expressed as



The equilibrium constant for the weak acid dissociation is
by $K_a = \frac{[H^+][A^-]}{[HA]}$ (ii)

Writing H^+ (ion concentration) subject of the formula)

$$H^+ = K_a \frac{[HA]}{[A^-]} \quad (iii) \text{ i.e. the } H^+ \text{ ion concentration}$$

Taking the negative log of equation (iii) we obtain a convenient expression for calculating a pH of buffer solution.
But $pH = -\log [H^+]$

$$pH = pK_a + \log \frac{[A^-]}{[HA]} \quad (iv)$$

Equation (iv) is known as the Henderson-Hasselbalch equation. This equation can be generated as

$$[pH = pK_a + \log \frac{\text{salt}}{\text{acid}}]$$

This equation is for buffers consisting of weak acids & its salt. It can be used to calculate the following.

- * The pH of a buffer solution
- * How much acid or salt is needed to make a buffer solution of a required pH.
- * The effect of the pH of a buffer solution when a small amount of acid or a base is added.

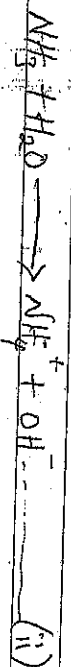
(ii) Weak Base Buffers

Buffer Solutions with constant pH values between 7 & 10

- Can be prepared from a weak base & one of its salt. A solution of NH_3 is typically used for this purpose. In aqueous solution, ammonium chloride dissociates into



NH_3 is a weak base, partially dissociates into



If acid is added, it is neutralized by hydroxide ions. As a result, equilibrium of equation (ii) shifts to the right. The shift maintains a constant concentration of hydroxide ions & thus a constant pH.

If a base is added, equilibrium of equation (ii) shifts to the left, thus maintaining the concentration of OH^- ion constant. The presence of NH_4Cl in the buffer solution ensures that there is sufficient NH_4^+ to cope with additions of base. The Henderson equation for a buffer solution made up of a weak base & one of its salt is given by

$$\text{for weak base } \text{pH} = \text{pK}_b + \log \left[\frac{\text{base}}{\text{salt}} \right]$$

(iii) Strong Base Buffer

A strong base can be used as a buffer solution with a pH value. The addition of small amount of acid or base has negligible effect on the pH e.g. when 1 cm^3 of 0.1 M of HCl is added to 100 cm^3 of 0.01 mol/dm^3 NaOH solution. The pH drops from 12 to 11.94. This drop of 0.04 can be checked using the equation for calculation of the pH of the base before and after the addition of the acid. For a strong base, to determine the pH we use the equation:

(iv) Strong Acid Buffer

A strong acid and as HNO_3 can act as a buffer with low pH. Strong acids are fully dissociated in aqueous solution. Thus, the concentration of H_3O^+ ions is high. The addition

(14)

- If small amt of acid or base will thus have a negligible effect on the pH of the acid. Eg if 1m^3 of HCl acid with concentration of 0.1mol/dm^3 added to 100m^3 of 0.01mol/dm^3 HCl , the pH decrease from 2 to 1.96. This negligible decrease of -0.04 can be checked using the pH equation to calculate the hydroxide ion concentration $\text{pH} = -\log [\text{H}^+]$ before & after the adding of the HCl .

Uses of Buffer Solution

Buffer Solutions plays an important role in many industrial & biological processes, and even those processes in human body. In industrial processes, processes such as electroplating, manufacturing of dyes, photographic materials, leather processing, calibrating of pH meters and wide varieties of chemical analysis employed the use of buffer solution. Buffer solutions are also used in biological research to maintain the pH of a culture media.

Problems involving the use of Buffer Solution

$$\text{pH} = \text{pK}_a + \log \frac{\text{salt}}{\text{acid}}$$

$$\text{pH} = \text{pK}_a - \text{pK}_b + \log \frac{\text{base}}{\text{acid}}$$

Example

1. How much Sodium ethanoate (CH_3COONa) must be dissolved in 1dm^3 of acetic acid (CH_3COOH) with concentration 0.01mol/dm^3 to produce a buffer solution of pH value of 5 & $\text{pK}_a = 4.75$

Answer

Data

$$\text{pK}_a = 4.75$$

$$\text{pH} = 5$$

$$\text{acid conc.} = 0.01\text{mol/dm}^3 \Rightarrow 10^{-2}$$

$$\text{salt conc.} = ?$$

$$\text{pH} = \text{pK}_a + \log \frac{\text{salt}}{\text{acid}}$$

Marking salt the subject of the formula

$$\log [\text{salt}] = \text{pH} - \text{pK}_a + \log [\text{acid}]$$

$$= 5 - 4.75 + (-2)$$

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$$= 0.25 - 2$$

$$\log [\text{salt}] = -1.75$$

finding the negative

$$\text{salt conc.} = -\log [1.75]$$

$$= 0.178 \approx 1.78 \times 10^{-2}$$

$$\text{No. of moles} = \frac{\text{mass}}{\text{molar mass}}$$

molar mass

$$\text{Gout molar mass of } \text{CH}_3\text{COOH} = 82 \text{ g mol}^{-1}$$

$$\text{If 1 mole of } \text{CH}_3\text{COOH} = 82 \text{ g mol}^{-1}$$

$$\text{then } 1.78 \times 10^{-2}$$

making X subject of the formula

$$X = 1.458 \approx 1.46 \text{ g}$$

Henderson Hasselbalch equation

For a buffer Addition of a weak acid & its conjugate base

$$\text{pH} = \text{pK}_a + \log \frac{[\text{conjugate base}]}{[\text{acid}]}$$

For a buffer Addition of a weak base & its conjugate acid

$$\text{pOH} = \text{pK}_a + \log \frac{[\text{conjugate base}]}{[\text{base}]}$$

Example.

Calculate the pH of a solution containing 0.20 mol dm^{-3} CH_3COOH & 0.30 mol dm^{-3} CH_3COONa (K_a of $\text{CH}_3\text{COOH} = 1.8 \times 10^{-5} \text{ mol dm}^{-3}$)

Answer:

For a buffer mixture of CH_3COOH & CH_3COO^-

$$\text{pH} = \text{pK}_a + \log \frac{[\text{conjugate base}]}{[\text{acid}]}$$

data

$$\text{conjugate base} \neq \text{CH}_3\text{COO}^- = 0.30 \text{ mol dm}^{-3}$$

$$\text{acid} (\text{CH}_3\text{COOH}) = 0.20 \text{ mol dm}^{-3}$$

$$\text{pH} = -\log (1.8 \times 10^{-5}) + \log \frac{[0.30]}{[0.20]}$$

$$= -(4.74) + 0.18$$

$$= 4.92 //$$

(16)

SURFACE PHENOMENA AND COLLOIDAL SYSTEM

Surface Phenomena & Colloidal System are closely related concepts in physical chemistry.

* Surface Phenomena

Surface Phenomena refers to the unique properties & behaviour that occur at the interface between two phases, such as:

1. Solid-liquid (eg. adsorption, wetting)
2. Liquid-gas (eg. surface tension, foam formation)
3. Solid-gas (eg. adsorption, catalysis)

These phenomena arise due to the difference in molecular interaction & arrangement at the interface compared to the bulk phases. Some key aspects of surface phenomena include.

* Colloidal System

A colloidal system also known as colloid is a mixture in which one substance is dispersed in another substance, but not dissolved. The dispersed substance (the colloid) has particles with diameters ranging from 1-1000 nanometers (nm). Examples of colloidal system include:

1. Suspensions (eg. mud, blood)
2. Emulsions (eg. mayonnaise, milk)
3. Foams (eg. soap foam, beer foam)
4. Aerosols (eg. fog, smoke).

→ Surface phenomena play a crucial role in colloidal systems. The interactions between the dispersed particles & the surrounding medium are influenced by surface phenomena, such as:

* Application of Colloidal System

A colloidal system is used as thickening agents in industrial product such as lubricants, lotions, toothpaste, coatings, etc. In the manufacture of paint & inks, colloids are useful. In ball point pens, the ink used is a gel (liquid-solid colloid).

* Colloidal gels: This is a type of gel-like substance that consist of a network of colloidal particles, trapping in the -

in the range of 1-1000 nanometers. dispersed in a solvent. These particles can be made of various materials, such as polymers, metals, or ceramics.

Colloidal gels have unique properties like, high water content; colloidal gels can contain large amount of solvent, making them useful in applications like cosmeceuticals, pharmaceuticals, & food products.

■ Surface tension is a physical property of liquids that cause them to behave as if they have an "elastic skin" at their surface. It's measured in units of force per unit length. typically in millinewtons per meter (mN/m).

Applications of Surface tension

1. Cleaning: Soap & detergent works by reducing surface tension making it easier to remove dirt & grime.
2. Medical applications: Surface tension plays a role in wound healing & the behavior of certain medical devices etc.

■ Adsorption is a process where one substance (e.g., a gas or liquid) penetrates into the bulk of another substance (e.g., a solid or liquid). This is different from absorption where the substance accumulates on the surface.

Adsorption is an important concept in various areas of chemistry, such as:

1. Adsorbents & Adsorbability.
2. Phase Equilibria.
3. Chemical Reactions.
4. Materials Science.

■ Emulsion; an emulsion is a mixture of two or more liquids that don't normally mix, such as oil & water. Emulsions are stabilized by adding an emulsifier, a substance that keeps the liquids mixed & stays together.

Characteristics of Emulsion

1. They are unstable mixtures; Emulsion are thermodynamically unstable meaning they will eventually separate into their individual components over time.
2. They stabilised by emulsifiers; emulsifiers, such as surfactants, lipids or polymers, help stabilising the emulsion by reducing the interfacial tension between the two liquids.
3. Microscopic droplets: Emulsions consist of tiny droplets of one liquid dispersed in another liquid.
4. Reversible or Irreversible: Emulsion can be reversible (e.g. mayonnaise) or irreversible (e.g. butter).

* Types of Emulsions

1. O/W - in-water (oil in water): oil droplets dispersed in water (e.g. milk, cream).
2. Water-in-oil (W/O): water droplets dispersed in oil (e.g. butter, margarine).
3. Multiple emulsions: Emulsions within emulsion (e.g. oil-in-water-in-oil).

* Applications of Emulsion

1. Emulsion are used in food products like mayonnaise and ice cream.
2. Emulsion are used in pharmaceuticals including delivery systems, such as creams & ointments.
3. They are also used in cosmetic product like lotions, creams & shampoos.
4. Emulsions are used in oil drilling & extraction process.

* Gels: A gel is a semi-solid substance that consists of a network of particles or polymer chains that are dispersed in a solvent. Gels are characterised by their viscoelastic properties.

Types of gels

1. Polymer gels
2. Colloidal gels
3. Hydrogels

14 Organogels

Applications of gels

1. Pharmaceuticals: gels are used as drug delivery systems, wound dressing, & implantable devices.
2. Cosmetics: gels are used in personal care products, such as hair gels, skin creams & toothpaste.
3. Food Industry: gels are used as thickeners, stabilizers, & texture modifiers in food products.

⇒ Some common examples of gels include

1. Hair gel.
2. Toothpaste.
3. Gelatin.

* Flotation is a process that separates solids from liquids by attaching gas bubbles to the solids. The bubbles cause the solids to rise to the surface, where they can be removed.

* Applications

1. Mineral Processing: Flotation is used to separate valuable minerals from rocks & soil from clays, silt & shale.
 2. Wastewater Treatment: Flotation is used to remove suspended matter.
 3. Concentrating Ores: Flotation is used to concentrate ores like copper, lead & zinc.
- ⇒ There are three types of flotation: natural, aided & induced flo-
tation. Flotation is the difference in density is naturally sufficient for separation.

* Chromatography

Chromatography is a laboratory technique used to separate, identify & quantify the components of a mixture. The word "chromatography" comes from the Greek words "chroma" (color) & "graphia" (writing), as the technique uses originally developed to separate & identify colored compounds.